Handling Data 6 — Drawing and interpreting histograms

Calculator allowed; graph paper needed. Marks in brackets. ★ = harder question. Frequency density $= \dfrac{\text{frequency}}{\text{class width}}$; area of a bar = frequency.

1 The masses of 120 apples are recorded in the table. [10]

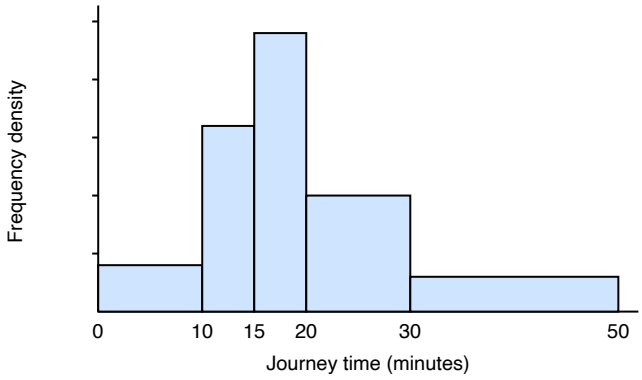
mass, m (g)	frequency
$60 < m \leq 80$	12
$80 < m \leq 90$	24
$90 < m \leq 95$	20
$95 < m \leq 100$	28
$100 < m \leq 110$	26
$110 < m \leq 130$	10

a Draw a histogram for this data.

b Estimate **i** the mean mass **ii** the median mass **iii** how many apples have a mass over 105 g.

c State the modal class.

2 The histogram shows the journey times to school of the students in a year group. 8 students took 10 minutes or less. [15]



a Work out the frequency density of the first class and hence label the vertical axis.

b How many students are in the year group? **c** Estimate the median journey time.

d Draw a frequency table for the data. **e** Estimate the mean journey time.

f Estimate how many students take longer than the mean time.

Total: 25 marks

Check: histogram heights are frequency densities, not frequencies · frequency = density × width (area) · mean: mid-points × frequencies, divide by total · median: find the class of the $\frac{n}{2}$th value, then go a fraction of the way through it · modal class = tallest bar · unequal widths → no bar may be drawn with its frequency as height.